



RIM-8 Talos

Bendix **RIM-8 Talos** was a long-range naval surface-to-air missile (SAM), among the earliest SAMs to equip United States Navy ships. The Talos used radar beam riding for guidance to the vicinity of its target, and semi-active radar homing (SARH) for terminal guidance. The four antennas surrounding the nose were SARH receivers, which functioned as a continuous wave interferometer. A solid rocket booster provided thrust for launch and a Bendix ramjet powered its flight to the target, with the warhead serving as the ramjet's compressor.

History

Talos was the end product of Operation Bumblebee, the Navy's 16-year surface-to-air missile development program for protection against guided anti-ship missiles like Henschel Hs 293 glide bombs, Fritz X, and kamikaze aircraft.^[1] The Talos was the primary effort behind the Bumblebee project but was not the first missile the program developed; the RIM-2 Terrier was the first to enter service. The Talos was originally designated SAM-N-6 and was redesignated RIM-8 in 1963. The airframe was manufactured by McDonnell Aircraft in St. Louis; final assembly was by Bendix Missile Systems in Mishawaka, Indiana. The first production versions of the missile cost about \$155,000 in 1955 (\$1,793,335 in 2022 dollars); however, the price would drop as Bendix increased production.^[2]

The Talos saw relatively limited use due to its large size and dual radar antenna system; few ships could accommodate the large missiles with the AN/SPW-2 missile guidance radar and the AN/SPG-49 target illumination and tracking radar.^[3] The 9.9-meter-long, 3½-ton

RIM-8 Talos



RIM-8G Talos missile

Type	Surface-to-air missile
Place of origin	United States
Service history	
In service	1958-1979
Used by	United States Navy
Production history	
Manufacturer	Bendix
Produced	1955
Specifications	
Mass	7,800 lb (3,500 kg) (missile: 3,400 lb (1,500 kg), booster: 4,400 lb (2,000 kg))
Length	32 ft (9.8 m)
Diameter	28 in (71 cm)
Wingspan	280 cm (110 in)
Warhead	211 kg (465 lb) continuous-rod HE warhead or W30 nuclear warhead (2–5 kt)
Engine	
Stage 1: Hercules MK 11 solid-fueled rocket booster, Stage 2: Bendix ramjet sustainer 20,053lbf, 89.20kN	
Operational range	RIM-8A: 92 km (50 nm); RIM-8J: 241 km (130 nm)
Flight ceiling	24,400 m (80,100 ft)

missile was comparable in size to a small fighter aircraft.^[4] The Talos Mark 7 Guided Missile Launching System (GMLS) was installed in three Galveston-class cruisers (converted Cleveland-class light cruisers) with 16 missiles in a ready-service magazine and up to 30 missiles and boosters in a storage area above the main deck. Nuclear-powered USS Long Beach and three Albany-class cruisers (converted Baltimore-class heavy cruisers) carried Mark 12 Guided Missile Launching Systems fed from a 52-round magazine below the main deck.^[5]

The initial SAM-N-6b/RIM-8A had an effective range of about 50 nmi and a conventional warhead. The SAM-N-6bW/RIM-8B was a RIM-8A with a nuclear warhead; terminal guidance was judged unnecessary for a nuclear warhead, so the SARH antenna was omitted. The SAM-N-6b1/RIM-8C was introduced in 1960 and had double the range and a more effective conventional continuous-rod warhead. The RIM-8D was the nuclear-warhead version of the -8C. The SAM-N-6c/RIM-8E "Unified Talos" had a warhead that could be swapped while embarked, eliminating the need to waste magazine capacity carrying dedicated nuclear-tipped variants. The RIM-8E also carried an improved continuous-wave terminal homing seeker and had a higher ceiling reach-out. Some RIM-8Cs were retrofitted with the new seeker and designated RIM-8F. The RIM-8G and RIM-8J had further radar homing improvements and a new fuel that extended the range to 130 nm.^[6]

The surface-to-air versions also saw action in Vietnam, with a total of four MiGs being shot down by USS Chicago and Long Beach. On May 23, 1968, a Talos fired from Long Beach shot down a Vietnamese MiG at a range of about 65 miles. This was the first downing of a hostile aircraft by a missile fired from a ship. The hit also destroyed a second MiG which flew through the debris. In September 1968, Long Beach scored another MiG destroyed at a range of 61 miles. On May 9, 1972, Chicago's forward Talos battery scored a long-range kill on a MiG.^[7] The Talos missile also had surface-to-surface capabilities.^[8]

The RGM-8H Talos-ARM was a dedicated anti-radar homing missile for use against shore-based radar stations. Initial testing of the RGM-8H was performed in 1965, and soon after, it was deployed in Vietnam on Chicago, Oklahoma City, and Long Beach, attacking North Vietnamese SAM radars. Oklahoma City fired the first successful RGM-8H combat shot in US Navy history in early 1972. It was also the first combat surface-to-surface missile shot in US Navy history.^[9]

Maximum speed	Mach 3
Guidance system	Radar <u>beam riding</u> and (non-nuclear variants) <u>semi-active radar homing</u>
Launch platform	Surface ship



Last Talos missile launched by USS Oklahoma City in 1979

Variants

SAM-N-6

Development and prototype missiles; pre-1962 US Navy designation of the Talos missile.

SAM-N-6a

Development and prototype missiles; pre-1962 US Navy designation of the Talos missile.

SAM-N-6b

Production missiles deployed with conventional explosive warheads; redesignated **RIM-8A**.

SAM-N-6bw

The -6b missile with a nuclear warhead, omitting terminal guidance and SARH antennas; redesignated **RIM-8B**.

SAM-N-6b1

An improved -6b with much greater range and continuous rod conventional warhead; redesignated **RIM-8C**.

SAM-N-6c

"Unified Talos" with interchangeable nuclear/conventional warheads eliminating the need for storage of both missile types, also fitted with improved terminal homing and higher operating ceiling; redesignated **RIM-8E**.

RIM-8F Talos

Some RIM-8C missiles retrofitted with the new seeker from the RIM-8E (post-1962 only).

RIM-8G Talos

Variant with further homing improvements.

RGM-8H Talos-ARM

A dedicated surface-to-surface anti-radar homing version for deployment on ships already fitted out for the Talos SAM.

RIM-8J Talos

Variant with further homing improvements.

MQM-8G Vandal

Talos missiles remaining after removal from active service were converted to super-sonic drone targets, with the inventory being exhausted circa 2008.

Chronology

Date	Fleet inventory	Ship	Event
28 May 1958 ^[10]	1 × Mk 7 GMLS 2 × AN/SPG-49 RADAR	<i>Galveston</i> ^[10]	Commissioned as CLG-3 ^[10]
3 June 1960 ^[11]	2 × Mk 7 GMLS 4 × AN/SPG-49 RADAR	<i>Little Rock</i> ^[11]	Commissioned as CLG-4 ^[11]
7 September 1960 ^[12]	3 × Mk 7 GMLS 6 × AN/SPG-49 RADAR	<i>Oklahoma City</i> ^[12]	Commissioned as CLG-5 ^[12]
9 September 1961 ^[13]	3 × Mk 7 GMLS 1 × Mk 12 GMLS 8 × AN/SPG-49 RADAR	<i>Long Beach</i> ^[13]	Commissioned as CGN-9 ^[13]
3 November 1962 ^[14]	3 × Mk 7 GMLS 3 × Mk 12 GMLS 12 × AN/SPG-49 RADAR	<i>Albany</i> ^[14]	Commissioned as CG-10 ^[14]
1 December 1962 ^[15]	3 × Mk 7 GMLS 5 × Mk 12 GMLS 16 × AN/SPG-49 RADAR	<i>Columbus</i>	Commissioned as CG-12
2 May 1964 ^[16]	3 × Mk 7 GMLS 7 × Mk 12 GMLS 20 × AN/SPG-49 RADAR	<i>Chicago</i> ^[16]	Commissioned as CG-11 ^[16]
25 May 1970 ^[17]	2 × Mk 7 GMLS 7 × Mk 12 GMLS 18 × AN/SPG-49 RADAR	<i>Galveston</i>	Decommissioned
31 January 1975 ^[15]	2 × Mk 7 GMLS 5 × Mk 12 GMLS 14 × AN/SPG-49 RADAR	<i>Columbus</i>	Decommissioned
22 November 1976 ^[18]	1 × Mk 7 GMLS 5 × Mk 12 GMLS 12 × AN/SPG-49 RADAR	<i>Little Rock</i>	Decommissioned ^[11]
1978	1 × Mk 7 GMLS 4 × Mk 12 GMLS 10 × AN/SPG-49 RADAR	<i>Long Beach</i>	Talos system removed
1 November 1979	4 × Mk 12 GMLS 8 × AN/SPG-49 RADAR	<i>Oklahoma City</i>	Last Talos fired
15 December 1979	4 × Mk 12 GMLS 8 × AN/SPG-49 RADAR	<i>Oklahoma City</i>	Decommissioned
1 March 1980 ^[16]	2 × Mk 12 GMLS 4 × AN/SPG-49 RADAR	<i>Chicago</i> ^[16]	Decommissioned ^[16]
29 August 1980 ^[14]		<i>Albany</i> ^[14]	Decommissioned ^[14]

Fate

Long Beach had her Talos launcher removed in 1978. Talos was phased out of fleet service with the decommissioning of *USS Oklahoma City* in 1979, though the *Albany*-class ships carrying the system soldiered on a few more years with the launchers left in place until they were retired in 1980. After 21

years of fleet service, the missile was replaced by the RIM-67 Standard missile, which was fired from the smaller Mk10 launcher.

Two Talos missiles are on display at the Military Honor Park located near the entrance of the South Bend International Airport in South Bend, Indiana.

A Talos missile was displayed in the atrium of the South Bend Regional Airport (historically known as Bendix Field), but was removed in 2021 to be displayed in the Manufacturing Victory exhibit at The History Museum in South Bend. ^[19] After the exhibit closed, the missile did not return to the airport.

Another example can be seen at the Patriots Point Naval & Maritime Museum, located at Mount Pleasant, South Carolina.

A Talos Missile can also be seen on display at the Muskogee War Memorial Park located in Muskogee, Oklahoma.

A Talos missile is on display at Naval Weapons Station Yorktown in Yorktown, Virginia.

A Talos missile is on display at The US Navy's Guided Missile School at Dam Neck, in Virginia Beach, Virginia, just outside of the main building of the NAVGMSCOL.

Two Talos missiles are on display, in launch position, on the stern of USS *Little Rock* at the Buffalo and Erie County Naval & Military Park located in Buffalo, New York.

A Talos missile and booster were on display at Rita Blanca Park (home of the XIT Rodeo & Reunion) in Dalhart, Texas, at least from 1981 or earlier, but as of 2017 had been removed.

A Talos missile is on display in the Steven F. Udvar-Hazy Center at the Washington Dulles International Airport.

A Talos missile is on outdoor display in front of the Missiles and More Museum on Topsail Island, NC. Notably, this location is also the birthplace of the RIM-8 Talos missile, having been a result of the research effort on ramjets and surface-to-air missiles which took place on Topsail Island from 1946 to 1948, as part of Operation Bumblebee.^[20]

Gallery



Talos missile launcher on USS *Long Beach*, July 1961



Talos missile guidance radars, AN/SPG-49.



RIM-8A and -8B missile launch.



A Talos shortly before hitting a B-17 target drone in 1957.



USS *Little Rock* fires a Talos, 4 May 1961.



Talos missiles on USS *Little Rock*, November 1960.



MQM-8G Vandal launch from San Nicolas Island, in 1999.



RIM-8 Talos missile loading conveyor aboard USS *Little Rock*.



RIM-8 Talos magazine racks in USS *Little Rock*



RIM-8 Talos display, Patriots Point Naval Museum, Charleston South Carolina



A Bendix RIM-8 Talos on display at the Steven F. Udvar-Hazy Center in Chantilly, VA

See also

- [RIM-2 Terrier](#)
- [RIM-24 Tartar](#)
- [Sea Dart](#), the [Royal Navy](#)'s similar design.

Notes

1. "A Brief History of White Sands Proving Ground 1941-1965" (<https://web.archive.org/web/20141028154222/http://nmsua.edu/tiopete/files/2008/12/wspgcoldbook.pdf>) (PDF). New Mexico State University. Archived from the original (<http://nmsua.edu/tiopete/files/2008/12/wspgcoldbook.pdf>) (PDF) on 2014-10-28. Retrieved 2010-08-19.
2. "RIM-8 Talos" (<https://www.globalsecurity.org/military/systems/munitions/rim-8.htm>). *Global Security*. Retrieved 1 November 2020.
3. Polmar, Norman (December 1978). "The U.S.Navy: Shipboard Radars". *United States Naval Institute Proceedings*.
4. The contemporary Soviet MiG-15 jet fighter was 10.1 meters long and weighed 5 tons.
5. Naval Training Command (1972). *Gunners Mate M 1&C* (NAVTRA 10200-B ed.). US Government Printing Office.
6. Garten Jr., William; Dean, Frank A. (April–June 1982). "Evolution of the Talos Missile". *Johns Hopkins APL Technical Digest*. **3** (2): 117–122. ISSN 0270-5214 (<https://search.worldcat.org/issn/0270-5214>).
7. "USS LITTLE ROCK CLG 4 / CG 4 TALOS MISSILE & MISSILE SIGHTINGS" (https://web.archive.org/web/20100802150545/http://www.ussslittlerock.org/Armament/Little_Rock_Talos_Missile.html). Archived from the original (http://www.ussslittlerock.org/Armament/Little_Rock_Talos_Missile.html) on 2010-08-02. Retrieved 2017-09-22.
8. "USS Oklahoma City - Talos Missile Firing Operations" (<http://www.okieboat.com/Talos%20firing%20operations.html>). Retrieved 2014-05-23.
9. Hays, Phillip R. "Details of the First Talos RGM-8H Anti Radiation Missile Combat Firing" (<http://www.okieboat.com/Talos%20antiradiation%20shot.html>). *USS Oklahoma City CL91 / CLG5 / CG5*. Retrieved 14 March 2017.
10. "Galveston II (CL-93)" (<https://www.history.navy.mil/research/histories/ship-histories/danfs/g/galveston-ii.html>). Naval History and Heritage Command. Retrieved 3 February 2022.
11. "Little Rock I (CL-92)" (<https://www.history.navy.mil/research/histories/ship-histories/danfs/l/little-rock.html>). Naval History and Heritage Command. Retrieved 3 February 2022.
12. "Oklahoma City I (CL-91)" (https://www.history.navy.mil/research/histories/ship-histories/danfs/o/oklahoma_city.html). Naval History and Heritage Command. Retrieved 3 February 2022.
13. "Long Beach III (CG (N)-9)" (<https://www.history.navy.mil/research/histories/ship-histories/danfs/l/long-beach-iii.html>). Naval History and Heritage Command. Retrieved 3 February 2022.
14. Doehring, Thoralf. "USS Albany (CG 10)" (<https://www.navysite.de/cg/cg10.htm>). *Unofficial US Navy Site*. Retrieved 3 February 2022.
15. "Welcome Aboard" (https://web.archive.org/web/20101102124521/http://www.usscolumbus.org/USS_Columbus/Welcome_Aboard.htm). USS Columbus Veterans Association. Archived from the original (http://www.usscolumbus.org/USS_Columbus/Welcome_Aboard.htm) on 2010-11-02. Retrieved 2010-08-27.
16. Yarnall, Paul L. "USS CHICAGO (CA 136 / CG 11)" (<http://www.navsource.net/archives/04/136/04136.htm>). *NavSource Online*. Retrieved 3 February 2022.
17. "Chronology - U.S.S. Galveston CL-93 / CLG-3" (<http://www.ussgalveston.org/archiveddocs/timeline.htm>). USS Galveston Shipmates Association. Retrieved 2010-08-27.
18. "A Brief History of the USS Little Rock" (<https://web.archive.org/web/20101223032403/http://ussslittlerock.org/Ship%20History.html>). USS Little Rock Association. Archived from the original (<http://www.ussslittlerock.org/Ship%20History.html>) on 2010-12-23. Retrieved 2010-08-27.
19. "South Bend airport's Talos missile taken to History Museum for manufacturing exhibit" (<https://www.southbendtribune.com/story/news/2021/06/16/cold-war-era-missile-moved-south-bend-airport-museum-exhibit/7709247002/>). *South Bend Tribune*. Retrieved 15 February 2024.
20. Missiles and More Museum. (2014, January 26). Operation Bumblebee. Missiles and More Museum. <https://missilesandmoremuseum.org/exhibits/operation-bumblebee/>

References

- Friedman, Norman (1982). "The "3 T" Programme". *Warship*. **VI** (22–3). London: Conway Maritime Press: 158–166, 181–185. ISBN 0-87021-981-2.
- Goss, Wilbur H.; et al. (April–June 1982). "Talos in Retrospect". *Johns Hopkins APL Technical Digest*. **3** (2): 116–179. ISSN 0270-5214 (<https://search.worldcat.org/issn/0270-5214>).

External links

- [Designation systems.net – RIM-8 Talos \(http://www.designation-systems.net/dusrm/m-8.html\)](http://www.designation-systems.net/dusrm/m-8.html)
 - [Talos missile and launching system, Talos history \(http://www.okieboat.com/Talos.html\)](http://www.okieboat.com/Talos.html)
 - [Talos Missile Handling, Cruiser Installation Film \(https://www.youtube.com/watch?v=laeDP4p1qZI\)](https://www.youtube.com/watch?v=laeDP4p1qZI)
-

Retrieved from "https://en.wikipedia.org/w/index.php?title=RIM-8_Talos&oldid=1296557043"